

Computer Networking: Concepts

(CSE 3751)

Experiment 4

Aim:

Implementation and analysis of inter-network communication in single router and dual router-based network configurations using Cisco Packet Tracer.

Objectives:

1. An overview on inter-network communication between multiple LANs using routers.
2. Configuration of multiple LANs using proper IPv4 addressing and establishment of communication between PCs in different networks through a single router.
3. Configuration of multiple LANs using proper IPv4 addressing and establishment of communication between PCs in different networks using two routers.
4. Verification of network connectivity using basic network diagnostic commands such as *ping* and *tracert*.

Exercises:

1. Explain how the router facilitates communication between two different LANs.
2. Write the purpose of the *ip route* command used in static routing.
3. What happens if the serial link between two routers fails? How can redundancy be implemented?
4. How can you verify whether routing is working properly using Packet Tracer commands?

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THEORY

Multiple LANs use different IP networks, so devices cannot communicate directly without a router.

• Role of Router

A router is a Layer 3 device that interconnects different IP networks and forwards packets based on the destination IP address.

• Default Gateway

Each LAN's PCs send traffic destined for other networks to the default gateway, which is the router interface connected to that LAN.

• Packet Forwarding Process

1. Router receives the packet from a LAN
2. Reads the destination IP
3. Looks up its routing table
4. Forwards the packet out through the correct interface.

• Inter-LAN Communication

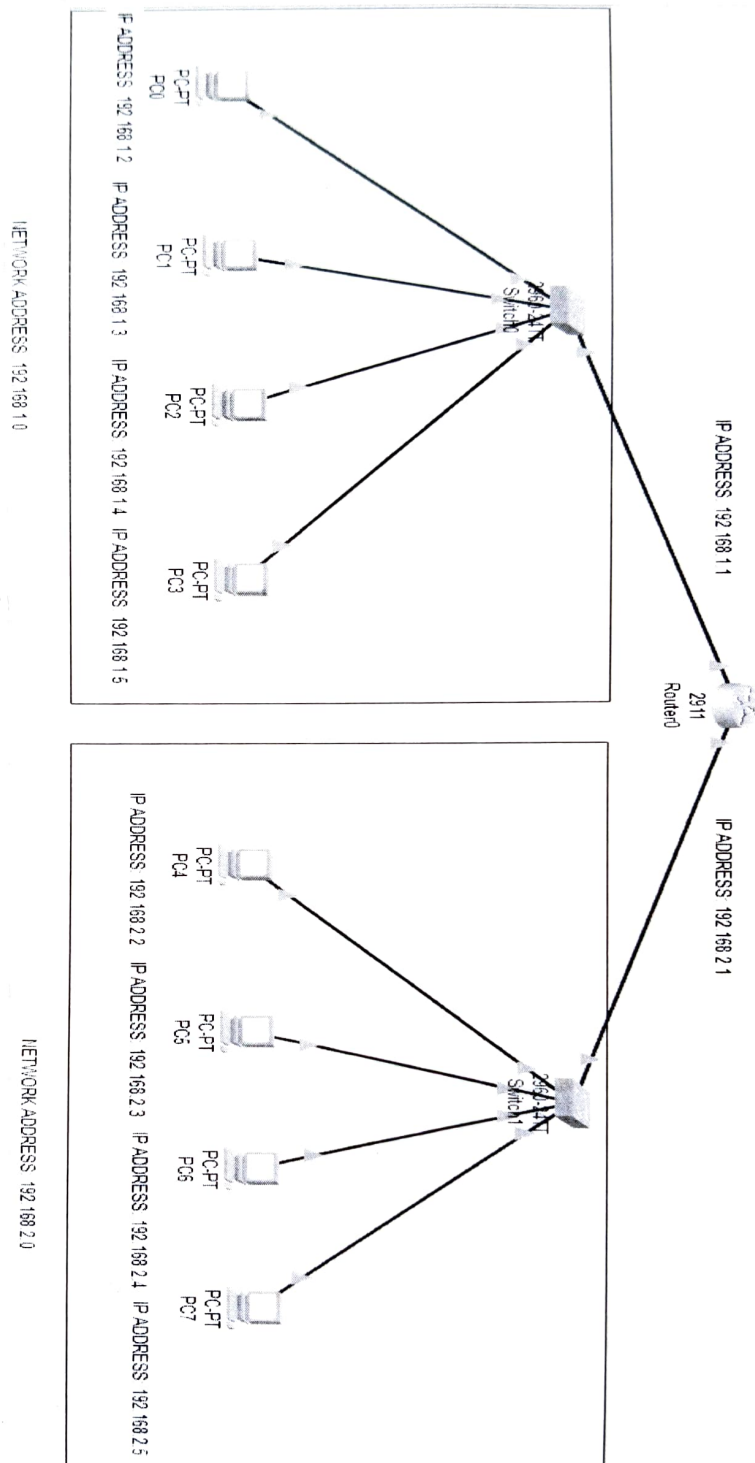
By connecting each LAN to a different router interface, the router enables communication between devices on different networks using IP routing.

So, Routers make inter-networks communication possible by providing a path between separate LANs, using IP addressing, routing tables, and default gateway forwarding.

P.T.O.

OBSERVATIONS

Fig. Connecting multiple LANs via single Router in Cisco Packet Tracer.
This is a clean implementation of basic inter-network routing. The diagram clearly demonstrates how a central 2911 router acts as the gateway to bridge 2 distinct subnets (192.168.1.0 and 192.168.2.0), allowing communication between the separate groups of PCs.



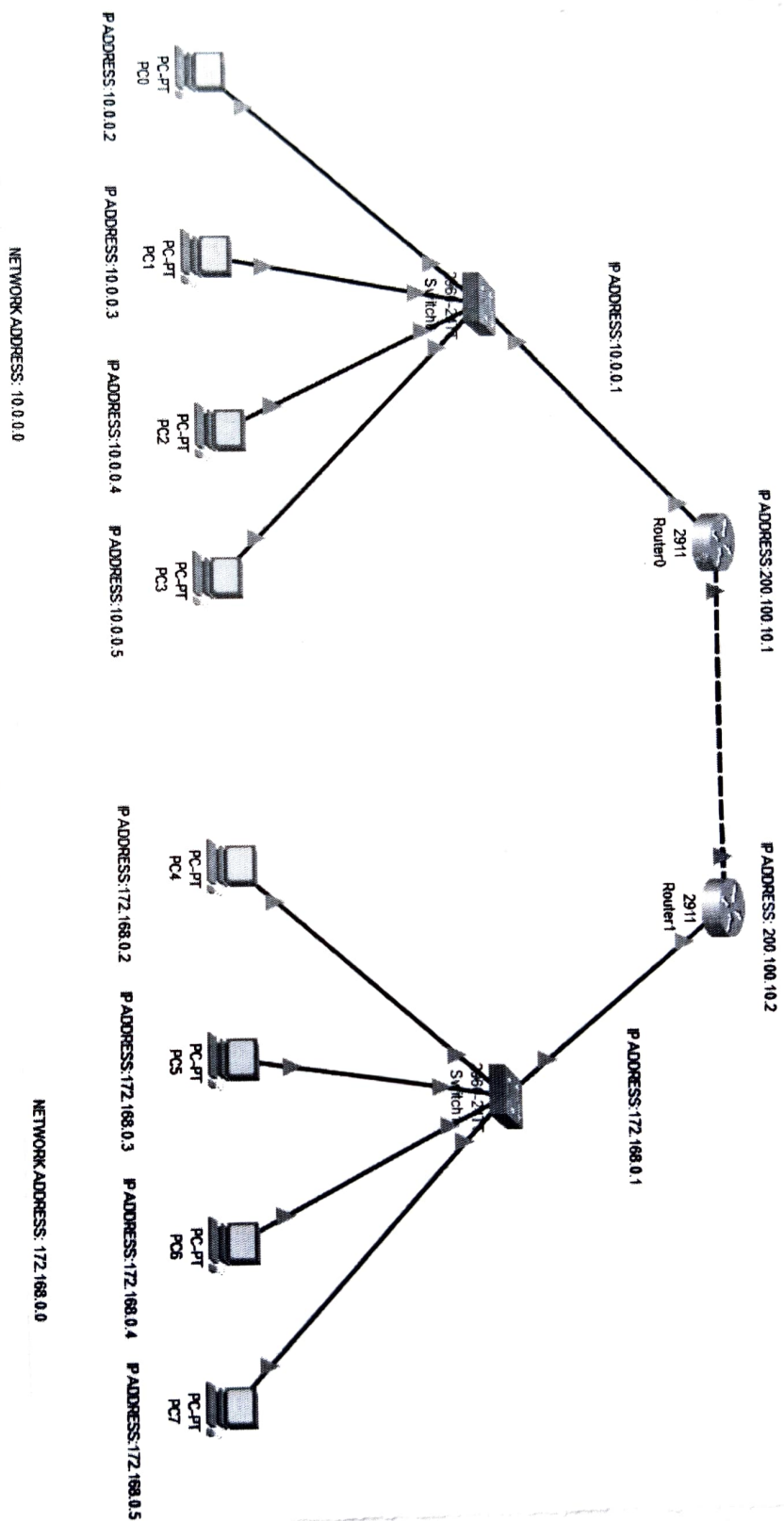
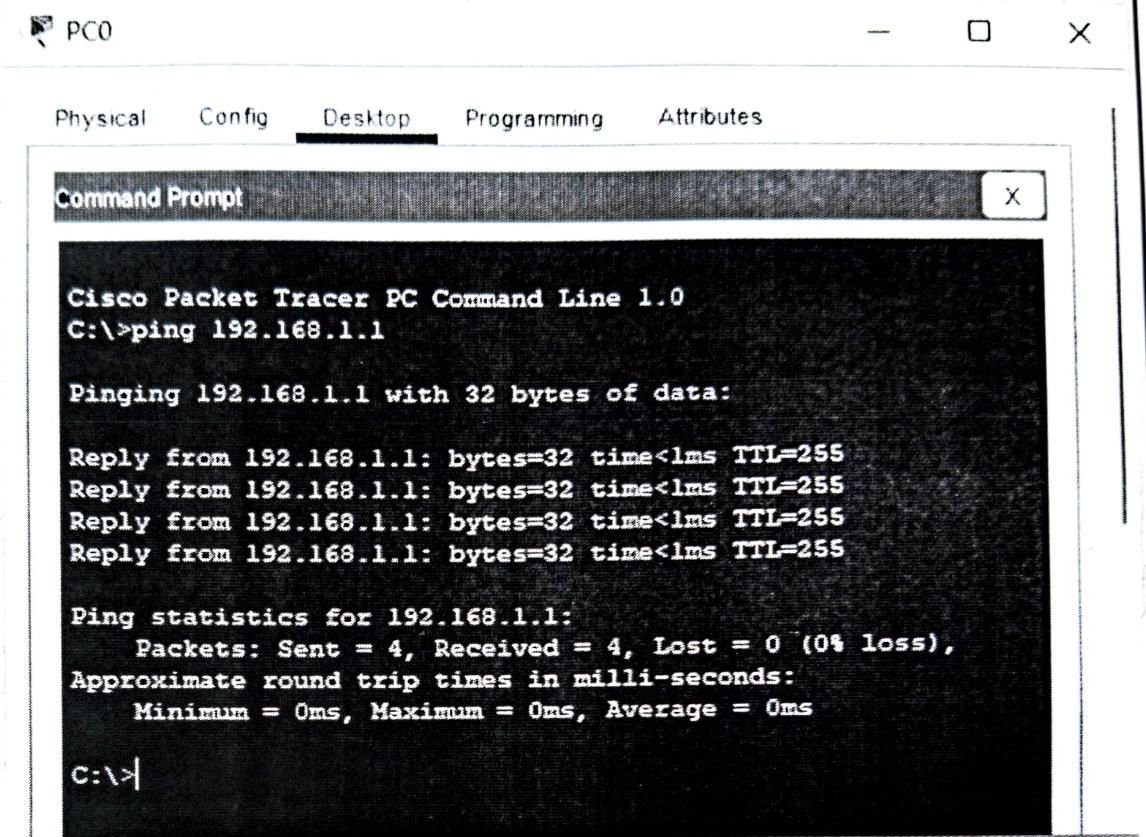


Fig. Multiple LANs connected via Dual-Router WAN Link.
This topology connects a Class A LAN (10.0.0.0) and a Class B LAN (172.168.0.0) via a dedicated WAN Link (200.100.10.2). Unlike a single router setup, this requires configuring static routing on RIP on both routers to allow data to travel between the two distinct networks.

• Verification of Multiple LAN using Single Router

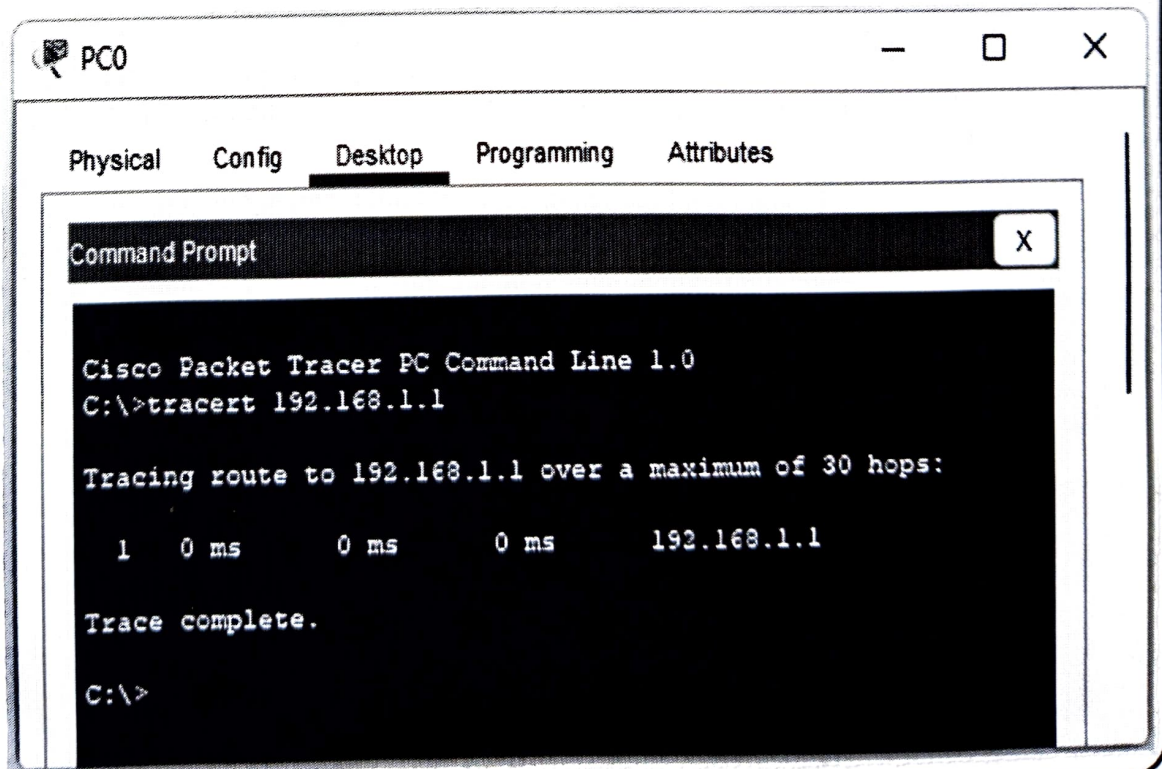
↳ ping command

Fig. Verifying local gateway connectivity.
PC0 successfully pings its default gateway (192.168.1.1),
confirming that the connection between the two devices and
the router interface is active and correctly configured on
the local subnet.



↳ tracert command

Fig. Route trace verification: Single-hop to Gateway.
The tracert command from PC0 confirms a direct, single-hop
connection to the router gateway (192.168.1.1) with
0 ms latency.



• Verification of Multiple LAN using 2 routers.

↳ using ping command.

Fig. Successful Routing Test (with ARP Delay)
The reply confirms that the data is successfully crossing both routers. The initial "Request timed out" messages are removed delays caused by the routers learning the addresses (ARP) for the first time.

PC0

```
C:\>ping 172.168.0.5

Pinging 172.168.0.5 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 172.168.0.5: bytes=32 time<1ms TTL=126
Reply from 172.168.0.5: bytes=32 time<1ms TTL=126

Ping statistics for 172.168.0.5:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

↳ using tracert command

Fig. Successful 3-Hop Route Verification.
The tracert output confirms the full path across the network.

PC0

```
C:\>tracert 172.168.0.5

Tracing route to 172.168.0.5 over a maximum of 30 hops:

  1  0 ms    0 ms    3 ms    10.0.0.1
  2  0 ms    0 ms    0 ms    200.100.10.2
  3  0 ms    0 ms    0 ms    172.168.0.5

Trace complete.
```

CONCLUSION

The experiment successfully demonstrated inter-network communication between multiple LANs using single and dual router setups. Proper IPv4 addressing was applied, routing was configured, and connectivity was verified using ping and tracert, meeting all objectives.

P.T.O.

ANALYSIS

Multiple LANs Using a Single Router.

In this configuration, two separate LANs (192.168.1.0 and 192.168.2.0) were connected to different interfaces of a singular router. The router acted as the default gateway for both networks (192.168.1.1 and 192.168.2.1). Since each LAN belongs to a different IP network, communication between them was possible only through the router.

The router received packets from one LAN, checked the destination IP, and forwarded the packet to the correct outgoing interface. This allowed PCs in 192.168.1.0 to successfully communicate with PCs in 192.168.2.0.

Ping Analysis

When a PC in 192.168.1.0 pinged 192.168.2.1, the reply was received successfully, showing 0% packet loss.

This confirms:

- Router interfaces are correctly configured.
- IP addressing is correct.
- Inter-LAN routing is functioning properly.

Tracert Analysis

The tracert output showed only one hop, indicating that the router directly routes packets between the two LANs without passing through any intermediate devices. This verifies that routing is operating at the local router level with no additional hops.

Multiple LANs Using Two Routers

In the dual-router setup, one LAN was connected to Router0 (10.0.0.0/24) and the other LAN was connected to Router1 (172.168.0.0/24). The routers were connected through a serial or intermediary link using the network 200.100.10.0/24. Static routes were configured on both routers so each router knew how to reach the other router's LAN. Once routing tables were configured correctly, devices from 10.0.0.0/24 and 172.168.0.0/24 were able to exchange packets.

Ping Analysis

- Ping succeeded with partial replies (at first).
- Then stable replies were received with low latency.

This confirms:

- Routing tables were updated correctly.
- The two routers were able to forward traffic to each other's LANs.
- Inter-network connectivity was established across routers.

Tracert Analysis

The tracert output showed three hops:

1. Hop 1 → The local ~~router~~ router (10.0.0.1)
2. Hop 2 → The remote router via 200.100.10.2
3. Hop 3 → Destination PC (172.168.0.5)

This proves that the packet travel from:

- Sender LAN → Router0 → Router1 → Destination LAN
- and confirms successful multi-router routing.

CONCLUSION

The experiment successfully demonstrated inter-network communication between multiple LANs using single and dual router setups. Proper IPv4 addressing was applied, routing was configured, and connectivity was verified using ping and traceret, meeting all objectives.

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EXERCISES

- (1) Explain how the router facilitates communication between two different LANs.
- (2) Write the purpose of the ip route command used in static routing.
- (3) What happens if the serial link between two routers fail? How can redundancy be implemented?
- (4) How can you verify whether routing is working properly using Packet Tracer commands?

Solutions

- (1) A router connects different LANs using its multiple interfaces, each assigned to a separate IP network. It receives packets from one LAN, checks the destination IP, looks up its routing table, and forwards the packet to the correct LAN. This enables communication between devices on different IP networks.
- (2) The ip route command is used to manually add routes to the router's routing table. It defines the next hop - address or exit interface for reaching a remote network, enabling communication between networks in static routing.
- (3) If the serial link fails, the routers cannot exchange packets, causing loss of connectivity between their LANs. Redundancy can be implemented by adding backup links, multiple paths, or using technologies like dual routers / alternate WAN connections so traffic can reroute automatically.
- (4) Routing can be verified using basic diagnostic commands:
- ping → tests connectivity to a destination network.
 - tracer / traceroute → shows the path packets take through routers.
- These commands confirm whether packets are correctly forwarded across networks.

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